Unified Implementation Plan for Autonomous CNC Toolpath System

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1 Included Source Prompts

1.1 Toolpath Prompt

You are a software+mechanical engineer tasked with generating a toolpath generation software... tool must enter from outside the face.

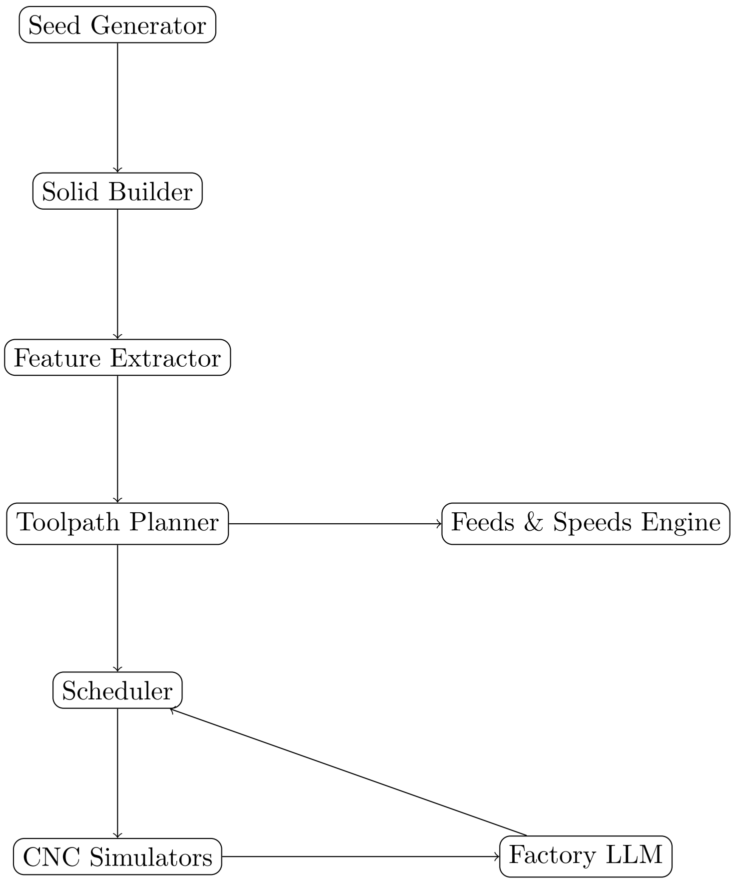
1.2 Feeds and Speeds Context

You are selecting feeds and speeds based on material, tool, and operation with shop-floor adjustments.

1.3 Factory Simulation Prompt

You are developing a simulation of an autonomous factory with multiple CNC machines, toolpaths, scheduling, and optimization policies.

2 System Architecture



3 Implementation Modules

3.1 1. Solid Builder

- Generate 1–4 cuboids (axis-aligned) - Add pockets and holes - Output: face list with wall/cliff metadata

3.2 2. Feature Extractor

- Identify top faces P$_i$ - Label edges: Wall / Cliff / Level

3.3 3. Toolpath Planner

Core algorithm:
- Classify edges (wall, cliff) [**?**]
- Compute safe region $S_i = P_i \ominus B_{blocked}$ [**?**]
- Generate raster paths - Ensure entry from safe edge

3.4 4. Feeds & Speeds Engine

- Compute RPM from cutting speed [**?**]
- Compute feed: $F = NZf_z$ [**?**] - Adjust based on chatter, rigidity

3.5 5. Drilling & Entry Module

- Peck drilling rules (0.25D–1D) [?] - Entry selection (plunge, ramp, helix)

3.6 6. Scheduler

- Assign parts to machines - Optimize based on policy:

- Min time
- Max tool life
- Min cost
- Best finish

3.7 7. CNC Simulator

- Execute G-code - Track:

- Tool wear
- Time
- Errors

4 Algorithm Pipeline

```
seed -> build geometry -> extract features
-> generate toolpath -> assign feeds/speeds
-> schedule -> simulate -> feedback
```

5 Key Data Structures

- Face: polygon + height + edges
- Tool: diameter, life, cost
- Machine: queue, tool crib
- Feature: pocket, hole, face

6 Weaknesses and Missing Information

6.1 Geometry Limitations

- Only cuboids supported - No curved surfaces or rotations

6.2 Toolpath Gaps

- No cutter deflection modeling - No corner smoothing/blending

6.3 Physics Gaps

- No cutting force model - No power/torque constraints

6.4 Feeds & Speeds Limitations

- Empirical only (no adaptive control) - Missing tool coating effects

6.5 Simulation Gaps

- Tool crib not fully implemented - Error handling incomplete - Multi-machine coordination unclear

6.6 Optimization Gaps

- Multi-objective optimization not formalized - No global optimization across features

6.7 Software Gaps

- Dependency on outdated OCC environment - No API contract between modules

7 Recommended Improvements

- Add force and power models
- Implement adaptive feed control
- Extend geometry kernel
- Formalize API between modules
- Add unit tests and benchmarking

8 Hierarchical Error-Aware Multi-Agent Control System

8.1 Overview

To address the current gap in error handling and multi-machine coordination, we introduce a hierarchical, LLM-driven control architecture enabling real-time anomaly interpretation, safe intervention, and coordinated mitigation across CNC agents and the factory system.

This system augments the existing CNC simulator and factory-level orchestration with:

- Interactive human-in-the-loop input
- Structured prompt-to-context transformation
- Local (CNC) and global (Factory) agent reasoning
- Safety-aware execution and escalation

8.2 System Architecture

The extended control loop is defined as:

`User Input` → `Context Transformer` → `CNC Agent LLM` → `Safety Evaluator` → `(Escalation)` → `Factory Agent LLM` → `Mitigation Execution`

8.3 CNC Agent Extensions

8.3.1 Simulation Dashboard

Each CNC simulator is extended with:

- Real-time toolpath and material removal visualization
- Machine telemetry (feed, RPM, load, tool state)
- **Interactive dialog box** for prompt input

The dialog box allows users to:

- Report anomalies (e.g., chatter, collision risk)
- Inject scenario-based conditions
- Directly interact with the simulation

8.3.2 Prompt-to-Context Transformation

Natural language prompts are transformed into structured representations:

```
{
  event_type: anomaly,
  category: vibration,
  severity: unknown,
  location: boundary_region,
  machine_state: {...}
}
```

This ensures that LLM reasoning is grounded in machine state rather than raw text.

8.3.3 CNC Agent LLM

The CNC agent LLM:

- Interprets structured context
- Proposes corrective actions

Example output:

```
{
  action: reduce_feed,
  parameters: {factor: 0.8},
  risk_level: moderate
}
```

8.3.4 Safety Evaluator

All LLM outputs are classified into:

- **Safe**: minor parameter adjustments
- **Risky**: significant modifications requiring monitoring
- **Catastrophic**: unsafe operations (e.g., collision risk)

8.3.5 Catastrophic Response Handling

If a catastrophic response is detected:

- Immediately pause machine execution
- Capture machine state (tool position, G-code index, telemetry)
- Escalate context to the factory agent

8.4 Factory Agent System

8.4.1 Factory Dashboard

A global interface providing:

- Multi-machine state visualization
- Production KPIs
- Interactive dialog interface

Supported commands include:

- Query machine state
- Stop production
- Modify scheduling and tooling policies

8.4.2 Factory Agent LLM

The factory agent:

- Performs system-level diagnosis
- Aggregates machine-level context
- Generates mitigation strategies

Example mitigation output:

```
{
  diagnosis: tool chatter near thin wall,
  actions: [
    {target: cnc_agent_5, command: reduce_feed, factor: 0.6},
    {target: scheduler, command: adjust_policy}
  ]
}
```

8.5 Multi-Agent Coordination

Agents communicate using structured messages:

```
{
  sender: factory_agent,
  receiver: cnc_agent_5,
  type: mitigation_command,
  payload: {...}
}
```

Coordination patterns include:

- Local correction (single CNC agent)
- Global policy update (scheduler or feeds/speeds)
- Multi-machine coordination

8.6 Integrated Execution Loop

The original pipeline:

`simulate` → `feedback`

is extended to:

`simulate` → `prompt` → `transform` → `CNC LLM` → `safety check` → `(escalate)` → `factory LLM` → `mitigate` → `resume`

8.7 Design Principles

- **Separation of Concerns**: CNC agent (local) vs Factory agent (global)
- **Structured Context**: no raw prompt reasoning
- **Safety First**: hard stop before unsafe execution
- **Human-in-the-Loop**: dialog interfaces enable supervision

8.8 Impact on Existing System

This extension directly addresses gaps identified in Section 6:

- Completes error handling architecture
- Enables multi-machine coordination
- Provides a foundation for adaptive and autonomous control

9 Conclusion

This unified plan integrates geometry generation, toolpath planning, machining parameters, and factory-level scheduling into a single modular system suitable for implementation.